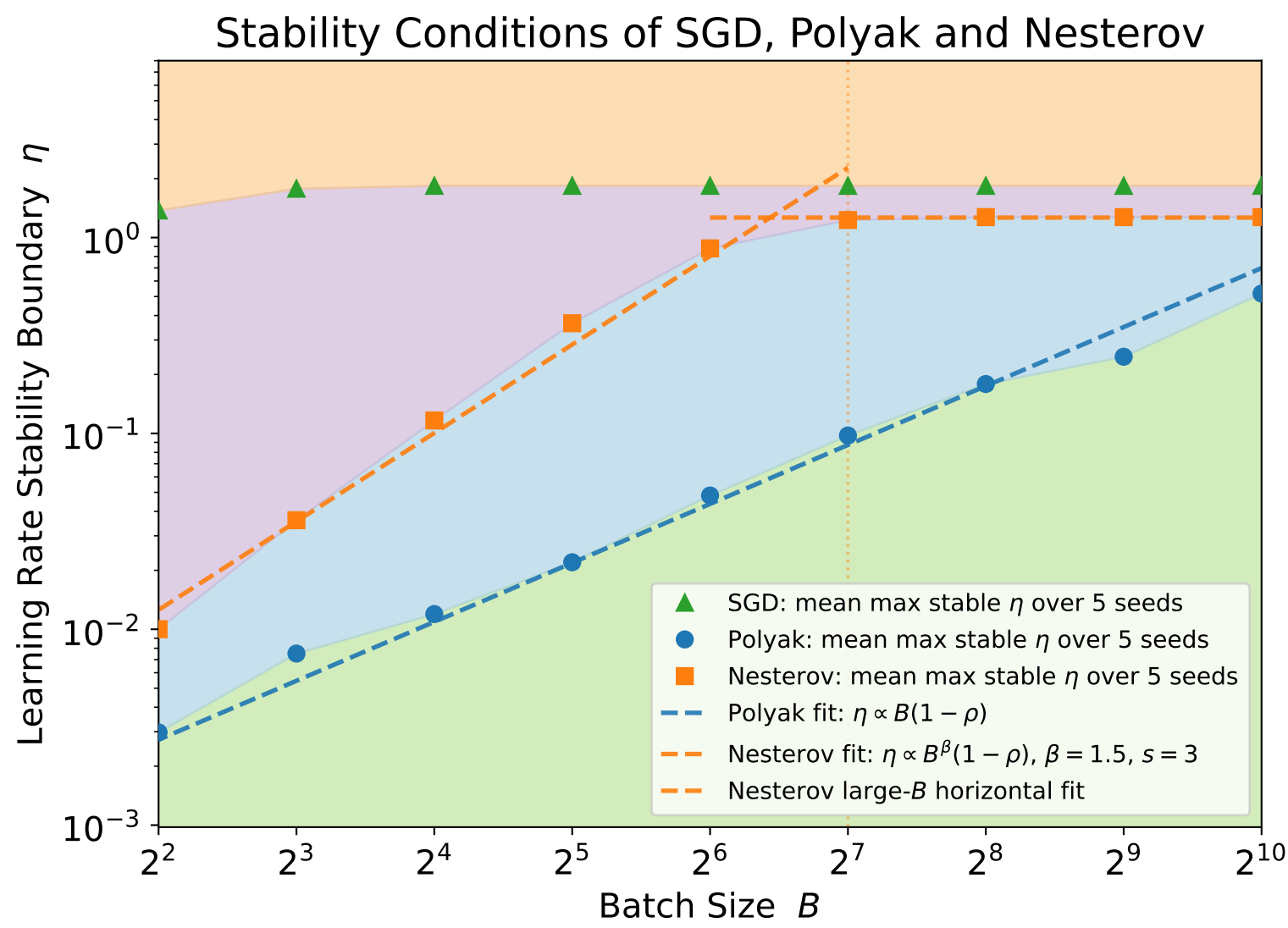
Stability Conditions of SGD, Polyak and Nesterov
Learning Rate Stability Boundary $\eta$
Batch Size $B$
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$2^{2}$
$2^{3}$
$2^{4}$
$2^{5}$
$2^{6}$
$2^{7}$
$2^{8}$
$2^{9}$
$2^{10}$
SGD: mean max stable $\eta$ over 5 seeds
Polyak: mean max stable $\eta$ over 5 seeds
Nesterov: mean max stable $\eta$ over 5 seeds
Polyak fit: $\eta \propto B(1-\rho)$
Nesterov fit: $\eta \propto B^{\beta}(1-\rho)$, $\beta = 1.5$, $s = 3$
Nesterov large-$B$ horizontal fit